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Is Diamond Open Access the Future of Open Access?

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Abstract

Open Access (OA) publishing is a publishing model that has arisen in recent years to combat the issues with traditional publishing. While traditional publishing models hide scholarly content behind expensive and prohibitive pay walls and subscriptions, OA seeks to provide access at little to no cost to users. While providing widespread access to academic content is a noble goal there has not been a consensus on how to achieve this goal. There are multiple OA publishing models but the most recent model is Diamond Open Access. The idea of Diamond OA is fascinating but it is difficult to understand; it produces high quality scholarly content without charging writers or users. It seems too good to be true and yet there are platforms, such as Episciences, which currently use this model. How these platforms function and whether or not this system can be more widely used in the future are topics of great interest. However, it is also important to examine this system carefully and not be distracted by its exciting ramifications. Diamond OA sounds very appealing at first glance; however, it is not without its flaws. This paper will present a balanced review in an attempt to answer the question: is Diamond OA the future of open access?

Keywords: Open Access, Diamond OA, scholarly publishing, publishing models, journals

In recent years, there has been a shift from a more traditional academic publishing model towards the Open Access (OA) model. The crux of the issue is that academic institutions essentially pay twice for scholarly content: they pay the researcher who produces the content and the publisher to access it. Increasingly this method of accessing scholarly content has become less viable as major publishers are charging more for their content. Open Access promotes the belief that the results of academic research should serve the public and that scholarly writing should be available for free. Open Access publishers produce academic content but utilize various methods to provide access to this content at

little to no cost to users. However, so far the OA publishing model has not provided much relief and large amounts of scholarly content remains behind expensive pay walls. OA, while it promises many benefits, does have its issues and faces many challenges. Three main OA models have come forward to try and address the problems with scholarly publishing: Green, Gold, and Diamond (sometimes called Platinum). These three models seek to make scholarly content accessible freely but have different ways of achieving this goal. Unfortunately, both Gold and Green OA require some form of payment; the Gold OA model requires some form of an author processing fee and the Green OA model requires payment if academic institutions want scholarly content immediately or if they want it edited and peer reviewed. Diamond OA has arisen as an alternative to these models that requires no payment on the part of the author or the institution. It is the newest OA model and is not yet well understood. While its benefits will be discussed in this paper it is difficult to see how the Diamond OA model can remain viable and provide a large scale solution to the issues within scholarly publishing.

This paper will examine the Diamond OA publishing model in depth. It will explain what Diamond OA is and assess whether or not it is feasible on a broad scale. First, this paper will compare the three OA publishing models to determine their strengths and weaknesses and to situate Diamond OA within this system. Second, it will look at examples of Diamond OA journals to see how they function. Finally, it will assess the feasibility of Diamond OA to determine if this model can be used long term and on a broader scale. Diamond OA sounds appealing at first glance; however, it is not perfect. This paper will present a balanced review in an attempt to answer the question: is Diamond OA the future of open access?

The OA movement started because of the problematic cycle of both producing and subsequently paying to access scholarly content created by traditional publishing models. Even in the digital age it costs money to publish scholarly content, and traditional publishers have argued that subscriptions and pay walls are necessary to remain sustainable. However, the major scholarly publishers charge significantly more than required to remain sustainable and reap massive profits off of the work of scholarly researchers. The onus of the system is on the academic institutions and scholars themselves. Researchers produce academic work as part of their jobs at academic institutions which is then given to publishers for free. Publishers once again rely on the free labour of academics to perform editing and peer review which is rarely compensated (Fuchs & Sandoval, 2013). Finally, the publishers charge subscription fees so academic institutions can access the scholarly content. As subscription fees increase many smaller institutions may find themselves unable to access important academic content (Koroso, *Diamond Open Access*, 2015; Suber, 2015).

This system remains viable due to the nature of academic careers. Publishing and performing the activities around publishing, like editing and peer review, are seen as important aspects of academic careers. Disseminating research is important to scholarly discourse and researchers want their work to be reviewed and circulated (Meadows & Wulf, 2016). Many academic institutions also consider the volume and type of publications of researchers in their tenure decisions (Fuchs & Sandoval, 2013).

However, this system would still function under the OA publishing models and as subscription fees become more prohibitive more institutions are turning to the different types of OA. There have also been arguments that research should be publically accessible especially if it was funded by government grants which are paid for by taxpayers (Koroso, *Diamond Open Access*, 2015). In fact, some research grants actually require the final product to be placed in an open access repository (Kelly, 2013). While it may be difficult to alter the traditional model, change does need to happen to ensure that scholarly content is accessible to those who need it.

The three OA publishing models all share the same goal: to provide free access to scholarly content. However, they approach this goal in different ways. The Green OA model still uses publishers who employ the traditional model to actually publish the work. However, with Green OA the author can also deposit the article into an institutional or discipline-specific repository that can be accessed freely by users (Esposito, 2016). The problem with this model is that the copy in the repository is often the unedited version that has not necessarily undergone peer review; to access the finalized version the academic institution still needs to pay the publisher (Kelly, 2013). There is also normally an embargo period so that the article only becomes available after this period is up. This means that the articles being accessed for free are not the best versions nor are they the most current research. Scholars also still depend on the traditional publishing model to have their work published which ultimately does little to stop the exploitative cycle of academic publishing. As a result, this model is not as widely used and while it does provide free access to scholarly content it seems it is done more as a way for publishers to pay lip service to open access.

Gold OA is the most common OA publishing model. There are different variations but they all revolve around the concept of the Author Processing Charge (APC), which is a fee the author pays so that the journal can cover the costs of publishing the article (Esposito, 2016). These fees can either be covered by the author themselves, or the institution can pay a membership fee so their scholars can publish in the journal (Kelly, 2013). There are also funding options, like grants, to cover these fees and encourage scholars to publish in OA journals. APCs allow scholarly content to be made available immediately and cover the costs of editing an peer review which helps improve the quality of content. Unfortunately, the Gold OA publishing model has several issues. The APCs can be very steep and thus prohibitive for some researchers who cannot secure the funds. Gold OA is also susceptible to abuse. Predatory publishers can target researchers by charging high fees but not providing the services promised for those fees. As a result, OA journals can have a reputation of being untrustworthy or not producing top quality academic content (Koroso, Open Access vs. Predator, 2015). These generalizations can harm the entire world of OA publishing. They discourage authors form publishing in OA journals and lead users to doubt the quality of the content.

Diamond OA was imagined as a way to circumvent these issues and ease concerns about the negative consequences of OA publishing (especially Gold OA). In the Diamond OA publishing model neither the author nor the institution pays for the costs of publishing the content; instead the funds

are provided from an outside donor. These donors can range from institutions, to governments, to private individuals or groups (Kelly, 2013). This means that the content published by Diamond OA journals still undergoes editing and peer review but the content remains free of cost. The Diamond OA model allows for high quality content to be made available freely and immediately.

While this model may seem like a fantasy, it is currently being used by a few platforms. *Episciences* is a hosting platform for scientific journals. It acts as both a repository and a publisher by producing overlay journals (Episciences, 2015). Researchers put their work into open archives which are available to everyone but the work has not undergone any editing or peer review. *Episciences* then selects content from this repository and produces overlay journals which are edited and reviewed by experts who are asked to contribute as part of editorial teams (Episciences, 2015). *Episciences* is funded and managed by le Centre pour la Communication Scientifique Directe (CCSD) and partners with open archives which are managed by multiple academic institutions. Another example of a Diamond OA platform is *Discrete Analysis* which is a mathematical journal. It too is an overlay journal but it does not create its own repository; instead it selects its content from arXiv, which is a repository hosted by Cornell University (Gowers, 2015). It has an editorial board of experts who select the content that will be reviewed and edited for the journal. *Discrete Analysis* uses Scholastica to host their content and provide access and while Scholastica charges $10 per submission, these fees are covered by grants from Cambridge University and the Stanhill Foundation (Gowers, 2015; Alliance of Diamond Open Access Journals, 2016). The *Croatian Medical Journal (CMJ)* is an interesting example of Diamond OA. Unlike the previously discussed platforms which are fairly recent, the *CMJ* is 25 years old and has successfully competed with journals published by traditional publishers for years (Gajović, 2017). The *CMJ* is run by 3 Croatian universities and receives funding from the Croatian Ministry of Science and Education. They receive submissions which are selected and edited by scholars from the controlling universities and reviewed by volunteers. All of their content is available on their website as well as on the PubMed Central repository (Gajović, 2017).

These examples clearly demonstrate the benefits of the Diamond OA publishing model. All their content is free to users as well as institutions and yet their content still undergoes the important steps to assure quality. In the case of the overlay journal model all of the stages of publishing are available to users, meaning that content can be accessed immediately as it is available as well as after it has been reviewed and edited. Since funding is secured from other sources Diamond OA journals are not as susceptible to the predatory publishing scams that can plague Gold OA journals. However, these benefits do not mean that Diamond OA is without its challenges. The biggest issue is that Diamond OA journals are completely dependent on outside funding sources. This is great for journals that can secure funding but this is obviously not feasible for every scholarly journal. Funding is also not guaranteed indefinitely. In the case of *Discrete Analysis*, the funding they received from Cambridge University is a fixed amount which will eventually be depleted, and they are currently searching for more funding to continue operations (Alliance of Diamond Open Access Journals, 2016). If funding

cannot be secured they will have to move to a Gold OA publishing model and charge a small fee so they can continue publishing. It is also possible that funding sources can add another layer of bias to the article selection process. If a donor does not want certain types of articles to be published it is not clear how much leverage they could have over this process. The Diamond OA model is also heavily reliant on volunteers to review, edit, and select the articles for publication. It is possible that non-financial incentives such as career benefits and prestige can attract volunteers but it is risky to rely completely on the good will of the community (Koroso, *Diamond Open Access*, 2015).

On the surface Diamond OA seems to be a good solution to the issues with open access. It provides the free access to quality content without the issues of how to pay for publishing. However, Diamond OA in its current form is not a good model for the future of open access. While it certainly has major benefits, it suffers from issues of long term feasibility and scalability. The reliance on external funding means that Diamond OA journals are inherently unstable. If the funding disappears the journal must resort to charging for publication or cease to exist. The example of the *CMJ* seems to prove that Diamond OA journals can remain viable long-term; however, their funding source is very stable government funding which is not a feasible option on a broad scale. When funding is available, Diamond OA provides everything one could want for open access but it is extremely improbable that even a small fraction of the thousands of scholarly publications will be able to secure adequate funding for long term operations. Therefore, while Diamond OA can provide access to a small number of high quality free journals it unfortunately cannot completely replace other publishing models. This is not to say that Diamond OA is doomed. A model similar to the one employed by *Episciences* or *Discrete Analysis* has the potential to deal with the limitations of Diamond OA. Overlay journals, which rely on submissions to a repository, reduce some of the work and costs associated with publishing. The other main costs covered by funding for Diamond OA are the costs of the hosting services used to provide access to the content. However, these costs can be relatively small. Scholastica, the hosting service used by *Discrete Analysis*, costs only $10 per submission - "…an amount roughly equal to the cost of a couple of beers…" (Gowers, 2015). Using the overlay journal model could keep APCs low which makes costs less prohibitive and reduces the temptation for predatory publishers. This is simply one example of how the lessons from Diamond OA can be used to progress open access. While Diamond OA in its current state may not be viable as a long term scalable solution, it does prove that open access without costs to the creators and users is possible. Diamond OA is relatively new and as it evolves it may become the ultimate solution to the issues with open access.

It is clear that the current publishing model employed by traditional publishers must change to better suit the needs of researchers and users. How this change will occur, unfortunately, is not clear. Even in the digital age publishing is not free and subscriptions and pay walls help cover the costs of publishing. However, as publishers seek to gain more profit, the academic institutions which provide and use the content are less and less able to access it. The open access movement was seen as a way to make scholarly content accessible to all, but the costs of publishing still need to be covered.

OA publishing models have proposed different solutions to this problem. Green OA provides access to articles through a repository but often they are unedited or reviewed and must first undergo an embargo period. Gold OA charges author processing charges to cover the costs of publishing but these costs can be prohibitively high and the system of charging authors to publish their research can be abused by dishonest publishers. Diamond OA charges neither the author nor the institution; the costs of publishing are covered by external funding sources. This model may seem to be the perfect solution; but as this paper has demonstrated, Diamond OA in its current state cannot fully replace existing publishing models. Diamond OA may not be the final solution to the challenges with OA publishing, but with each new model proposed a solution seems closer than ever.

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